

LINUX SYSTEM SECURITY AND NETWORK MONITORING PROJECT REPORT

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1. Introduction

Basic Linux system security and network monitoring using Kali Linux.

2. Objectives

To understand basic Linux network configuration
To identify the system IP address
To test network connectivity
To view network routes
To identify listening network ports
To verify the current Linux user

3. Tools and Technologies Used

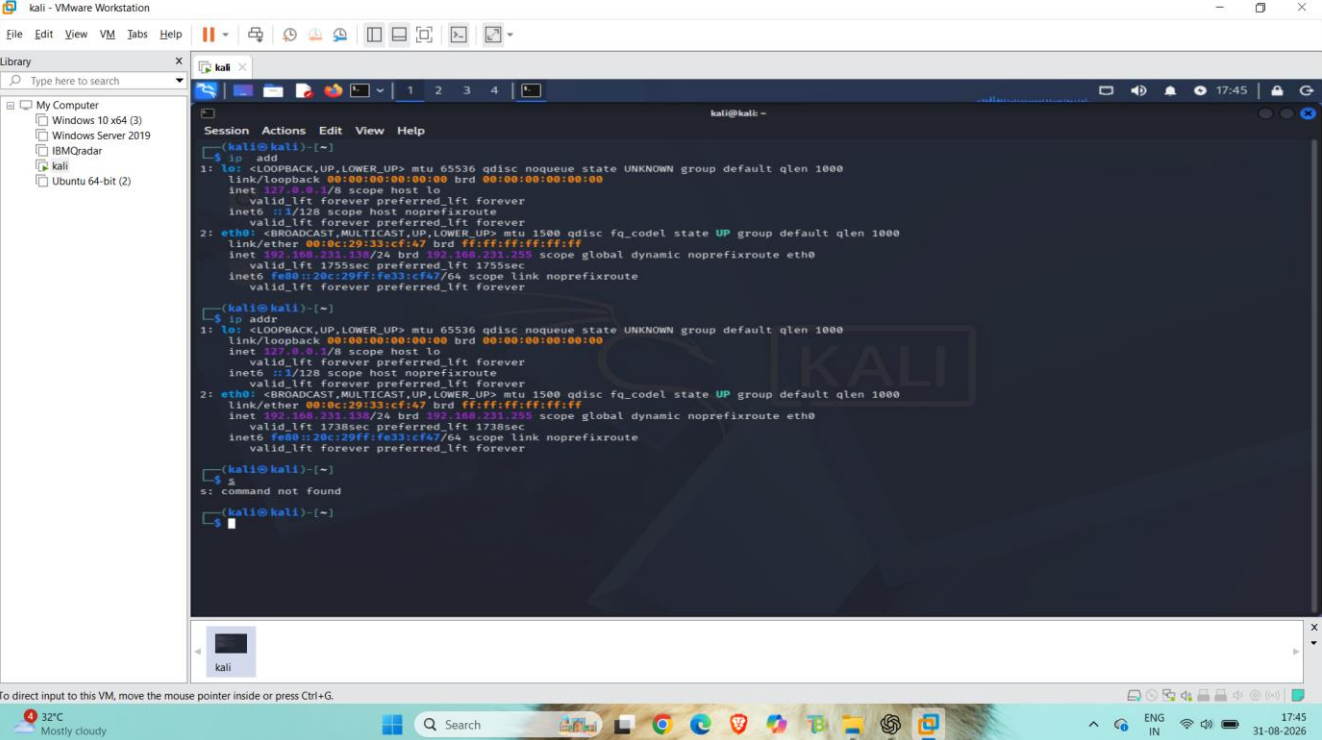
Kali Linux

Linux Terminal

4. IP Address Configuration

Command Used: ip addr

Screenshot 1 – IP Address



The screenshot shows a Kali Linux terminal window within a VMware Workstation environment. The terminal displays the output of the 'ip addr' command, which shows the configuration for the loopback interface 'lo' and the ethernet interface 'eth0'. The 'lo' interface is configured with the IP address 127.0.0.1 and the MAC address 00:00:00:00:00:00. The 'eth0' interface is configured with the IP address 192.168.231.139 and the MAC address 00:0c:29:33:c1:47. The terminal also shows the output of the 'ip netns exec' command, which is used to execute the 'ip addr' command within a specific network namespace.

```
kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Windows 10 x64 (3)
Windows Server 2019
IBMQradar
kali
Ubuntu 64-bit (2)

kali@kali:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:33:c1:47 brd ff:ff:ff:ff:ff:ff
    inet 192.168.231.139/24 brd 192.168.231.255 scope global dynamic noprefixroute eth0
        valid_lft 1755sec preferred_lft 1755sec
    inet6 fe80::29c:29ff:fe33:c147/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

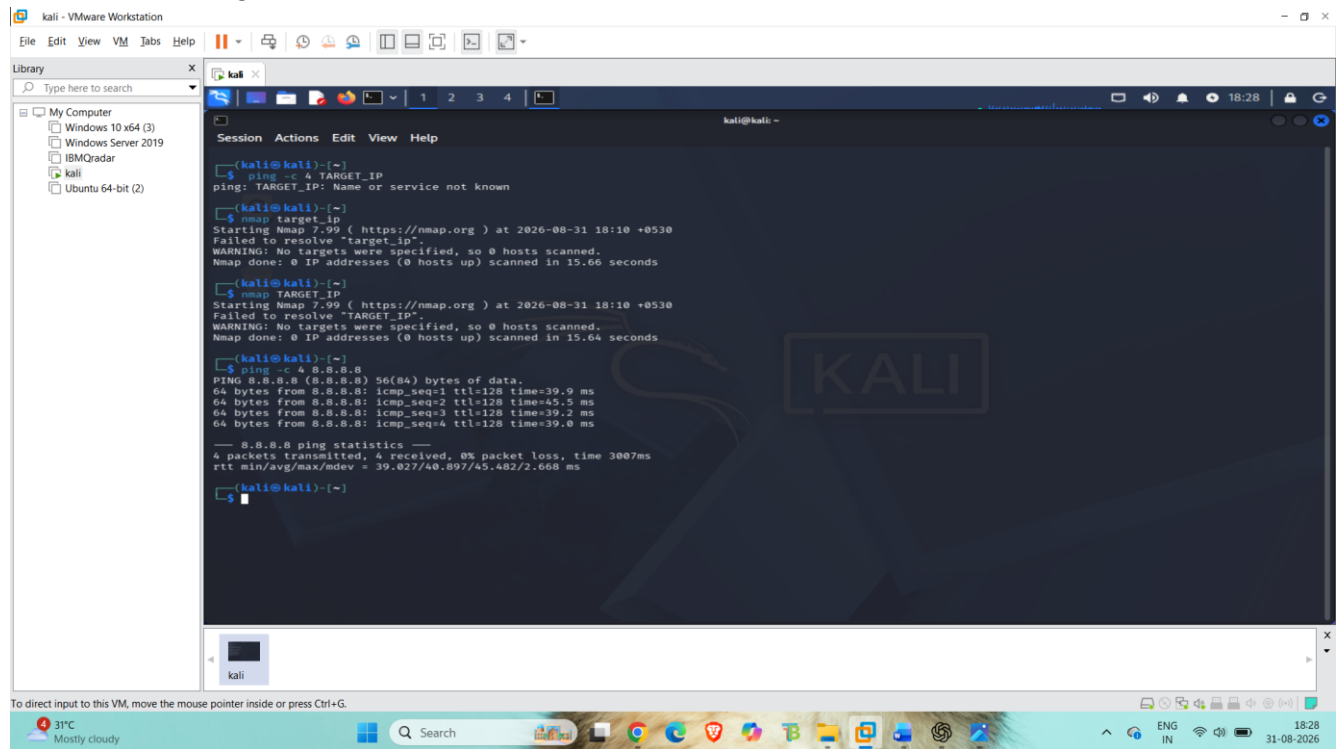
kali@kali:~$ ip netns exec
s: command not found

kali@kali:~$
```

5. Network Connectivity Testing

Command Used: `ping -c 4 8.8.8.8`

Screenshot 2 – Ping Test



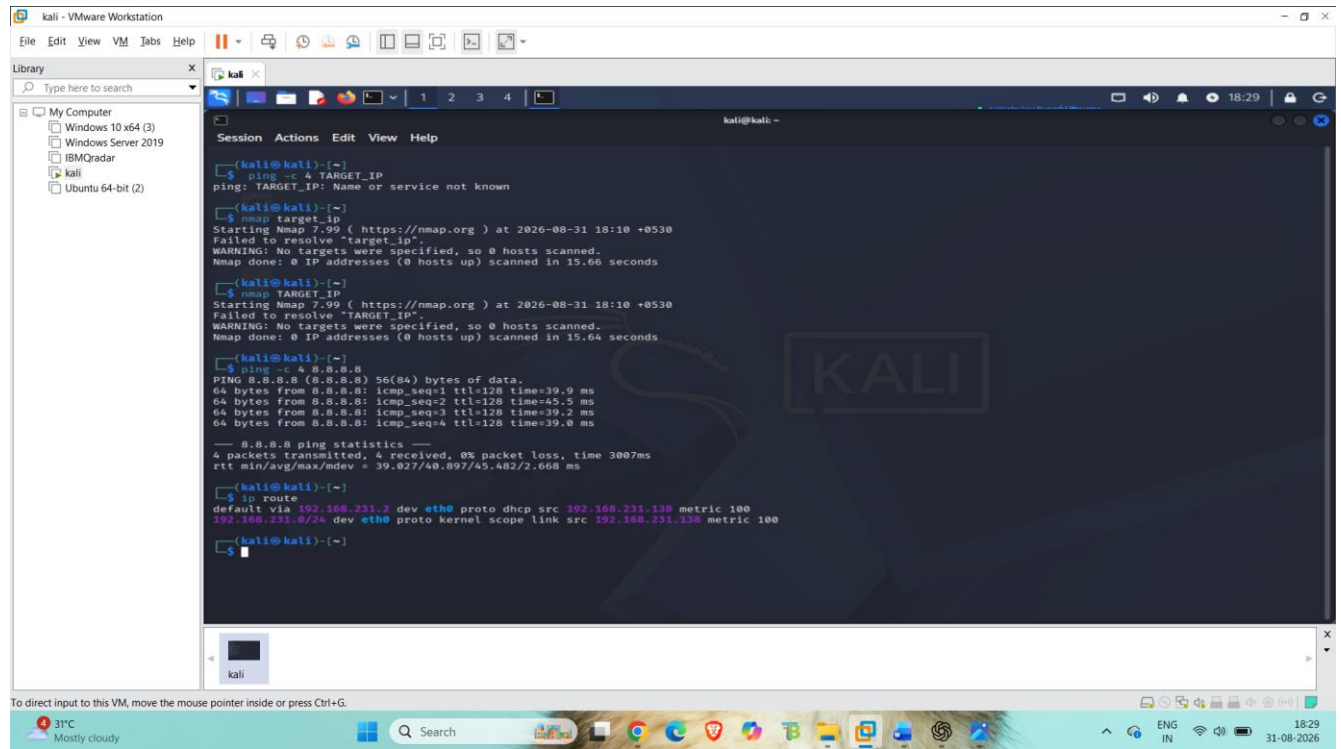
```
kali - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Windows 10 x64 (3)
Windows Server 2019
IBM Qradar
kali
Ubuntu 64-bit (2)

kali@kali:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=39.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=45.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=39.2 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=39.0 ms
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 39.027/40.897/45.482/2.668 ms
kali@kali:~$ nmap target_ip
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-31 18:10 +0530
Failed to resolve "target_ip".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 15.66 seconds
kali@kali:~$ nmap TARGET_IP
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-31 18:10 +0530
Failed to resolve "TARGET_IP".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 15.64 seconds
```

6. Network Route Information

Command Used: `ip route`

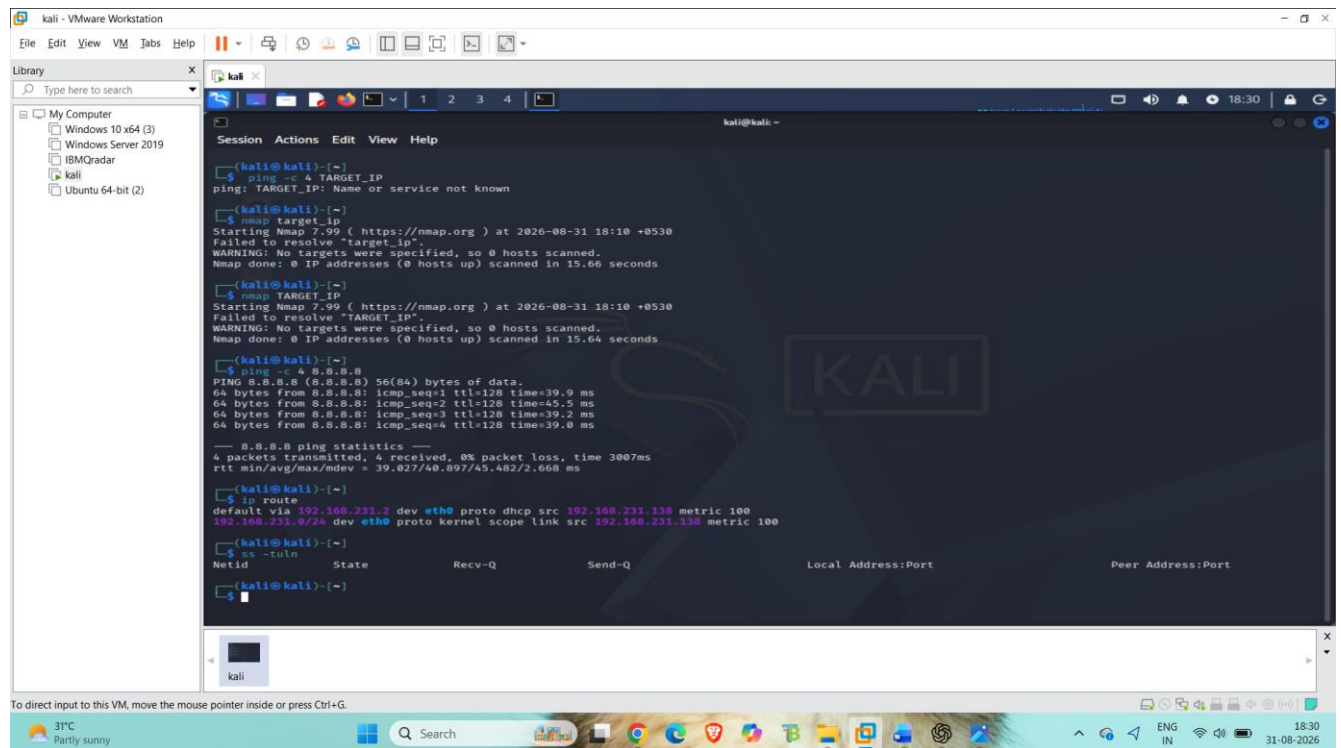
Screenshot 3 – Routing Information



7. Listening Network Ports

Command Used: ss -tuln

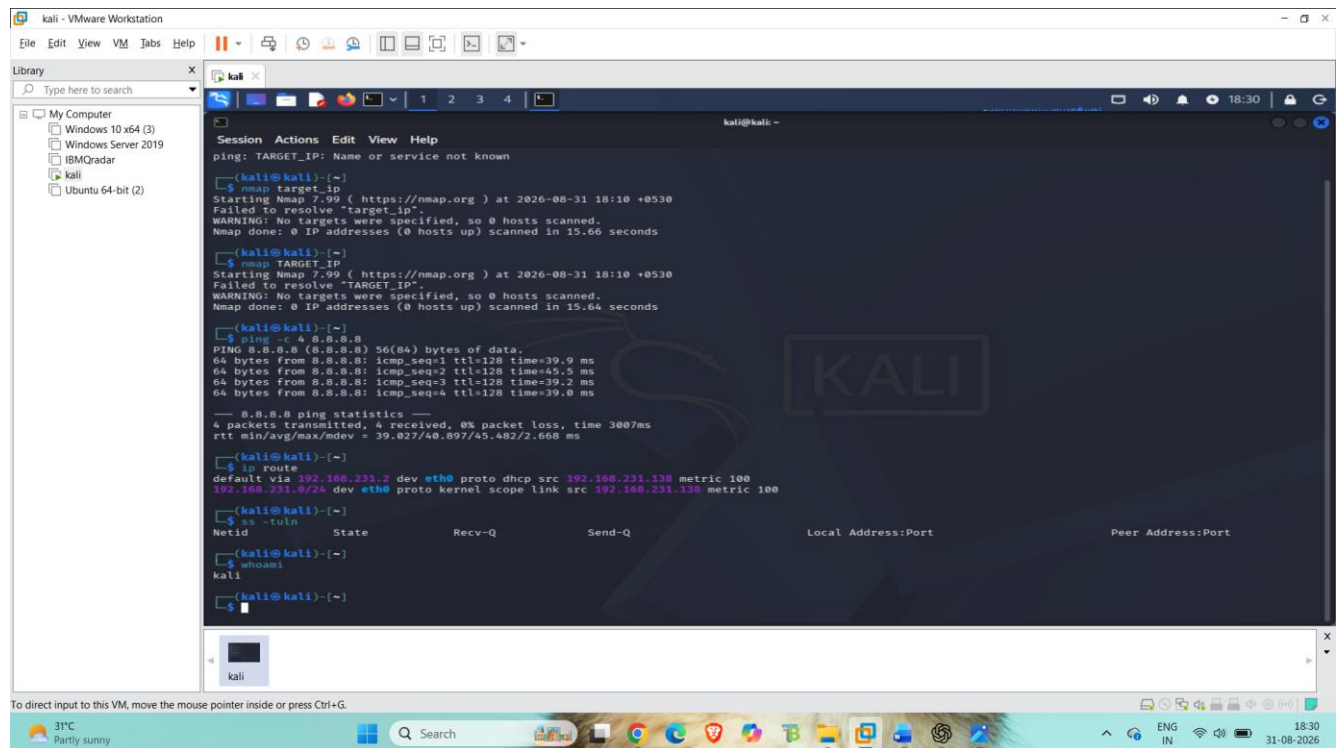
Screenshot 4 – Listening Ports



8. Current User Verification

Command Used: whoami

Screenshot 5 – Current User



9. Results

The system network configuration, connectivity, routing information, listening ports, and current user were successfully checked using Linux commands.

10. Security Recommendations

Monitor listening ports regularly

Disable unnecessary services

Use strong passwords

Keep the system updated

Allow only required network communication